This is a request for filing a PROVISIONAL APPLICATION FOR PATENT under 37 CFR 1.53(c).

INVENTOR(S):

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TITLE OF THE INVENTION (500 characters max):

NEXUS - Unified Card Management System with AI-Powered Deck Optimization, Hardware Scanning, and Business Intelligence for Trading Card Games

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ENCLOSED APPLICATION PARTS (check all that apply):

☒ Specification Number of Pages: 45

☒ Drawing(s) Number of Sheets: 8

☐ CD(s), Number of CDs: __

☐ Other (specify): _____

METHOD OF PAYMENT (check one):

☐ A check or money order is enclosed to cover the filing fees

☐ Payment by credit card (Form PTO-2038 attached)

☐ Charge Deposit Account [Account Number]

The Commissioner is hereby authorized to charge filing fees or credit any overpayment to Deposit Account Number: _____

Applicant claims small entity status. See 37 CFR 1.27

SIGNATURE:

_____

Kevin Caracozza

Date: November 21, 2025

TYPED OR PRINTED NAME: Kevin Caracozza

REGISTRATION NO. (if applicable): N/A (Pro Se)

PROVISIONAL PATENT APPLICATION

TITLE: NEXUS - Unified Card Management System with AI-Powered Deck Optimization, Hardware Scanning, and Business Intelligence for Trading Card Games

INVENTOR: Kevin Caracozza, 321 Lyndhurst Avenue, Lyndhurst, NJ 07071

FIELD OF THE INVENTION

This invention relates generally to collectible card game management systems, and more particularly to an integrated software and hardware system for automated card recognition, inventory management, AI-powered deck construction, and business intelligence analytics for trading card games such as Magic: The Gathering.

BACKGROUND OF THE INVENTION

Trading card games, particularly Magic: The Gathering, involve managing large collections of physical cards (often thousands to tens of thousands of individual cards). Existing solutions suffer from several critical limitations:

1. Manual inventory management is time-consuming and error-prone
2. Card recognition systems lack accuracy and require excessive user confirmation
3. Deck building tools don't integrate with actual inventory ownership

4. No unified system combines physical hardware scanning with digital inventory management
5. Business intelligence tools for card shop operations are disconnected from inventory systems
6. Market pricing data requires manual lookup across multiple platforms
7. No automated system exists for tracking card acquisition and disposition
8. Deck optimization lacks AI-powered analysis of card synergies and meta-game trends

Current market solutions include:

- Manual spreadsheet tracking (labor-intensive, error-prone)
- Mobile apps with camera recognition (low accuracy, no hardware integration)
- Online inventory tools (no physical scanning, manual data entry)
- Deck building websites (no inventory integration, no AI optimization)

None of these solutions provide an integrated hardware/software system that combines automated physical card scanning, AI-powered deck optimization, real-time market pricing, business intelligence analytics, and comprehensive inventory management in a single unified platform.

SUMMARY OF THE INVENTION

The present invention addresses these limitations by providing NEXUS - a unified system integrating:

1. HARDWARE SCANNING SUBSYSTEM:

- Arduino-controlled automated card positioning
- Dual camera system (DSLR + webcam) with DirectShow backend
- NeoPixel RGB LED lighting with 8-bit color control (0-255 per channel)
- Programmable servo motor for precise card positioning
- USB serial communication protocol (9600 baud)

2. AI-POWERED DECK BUILDING SUBSYSTEM:
- NumPy-based batch processing for performance optimization
- Multi-format support (Commander, Standard, Modern, Pioneer, Legacy, Vintage, Pauper, Brawl)
- Strategy-based deck construction (balanced, aggro, control, combo, midrange, tempo)
- Real-time mana curve analysis and color distribution optimization
- Card synergy detection using graph-based algorithms
- Meta-game analysis integration
- Collection-aware substitution suggestions

3. CARD RECOGNITION SUBSYSTEM:
- OCR-based text extraction from card images
- Fuzzy string matching against comprehensive card database (100,000+ cards)
- Scryfall API integration for card data validation
- Recognition confidence scoring with user confirmation workflow
- Batch processing mode for high-volume scanning
- Duplicate detection and consolidation

4. INVENTORY MANAGEMENT SUBSYSTEM:

- Real-time inventory tracking with quantity management
- Foil/hologram variant differentiation
- Set and edition tracking
- Card condition grading (NM, LP, MP, HP, DMG)
- Acquisition tracking (date, source, cost)
- Disposition tracking (sales, trades, deck building)
- CSV import/export for data portability
- Dewey Decimal-inspired organizational system

5. BUSINESS INTELLIGENCE SUBSYSTEM:

- QuickBooks CSV integration for financial tracking
- Customer ROI analytics (cost vs. revenue per customer)
- Sales forecasting using statistical models
- Trend analysis for card price movements
- Inventory turnover rate calculation
- Profit margin analysis by card, set, and category
- Market intelligence dashboard with real-time pricing

6. MARKETPLACE SUBSYSTEM:

- Internal peer-to-peer trading platform
- User reputation system with transaction ratings
- Watchlist functionality for tracking desired cards
- Offer negotiation system
- Transaction history and analytics
- Automated pricing recommendations based on market data

7. DECK TESTING SUBSYSTEM:

- Goldfish simulation (1000+ game iterations)
- Combat simulation against meta archetypes
- Mulligan decision optimization
- Win rate analysis and statistical reporting
- Mana base analysis and recommendations

The system architecture employs:

- Python 3.14 as primary programming language
- Tkinter with Gothic-themed UI (Innistrad/Ravnica aesthetic)
- NumPy 2.2.6 for numerical processing and batch operations
- OpenCV for image processing
- PIL/Pillow for image manipulation
- ThreadPoolExecutor for concurrent operations
- Serial communication (PySerial) for Arduino integration
- RESTful API integration with external services (Scryfall, TCGPlayer)

DETAILED DESCRIPTION OF THE INVENTION

1. HARDWARE ARCHITECTURE:

The physical scanning apparatus comprises:

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HARDWARE ARCHITECTURE - MECHANICAL POSITIONING SYSTEM

3. X-Y AXIS POSITIONING SYSTEM

The automated card scanning apparatus incorporates a precision X-Y axis positioning system for accurate card location and capture:

3.1 MECHANICAL COMPONENTS:

- X-AXIS LEAD SCREW ASSEMBLY:
  - Dual parallel lead screws (8mm diameter, 2mm pitch)
  - Length: 400mm travel distance
  - NEMA 17 stepper motor (200 steps/revolution)
  - GT2 timing belt drive system
  - Linear guide rails with ball bearing carriages
  - End-stop limit switches for homing
  - Positional accuracy: ±0.1mm

- Y-AXIS LEAD SCREW ASSEMBLY:
  - Single lead screw (8mm diameter, 2mm pitch)
  - Length: 600mm travel distance
  - NEMA 17 stepper motor (200 steps/revolution)
  - Direct drive configuration
  - Linear guide rails with ball bearing carriages
  - End-stop limit switches for homing

- Positional accuracy: ±0.1mm

• CAMERA CARRIAGE:

- Aluminum mounting plate (150mm x 100mm)
- USB camera mount with adjustable Z-height
- NeoPixel LED ring mount (integrated lighting)
- Cable management system (prevents tangling)
- Weight: <500g for minimal motor load

3.2 MOTION CONTROL SYSTEM:

• STEPPER MOTOR DRIVERS:

- A4988 or TMC2208 stepper drivers
- Microstepping capability (1/16 or 1/32)
- Current limiting for motor protection
- Integrated to Arduino Mega 2560 via RAMPS 1.4 shield

• POSITIONING ALGORITHM:

- Grid-based card layout (9x5 standard binder page)
- Automated home sequence on startup
- Sequential scanning pattern (left-to-right, top-to-bottom)
- Configurable card spacing parameters
- Collision detection via limit switches

• MOVEMENT SPECIFICATIONS:

- Maximum speed: 100mm/s

- Acceleration: 500mm/s^2
- Positioning time: <2 seconds per card
- Repeatability: ±0.05mm
- Backlash compensation: Software-implemented

3.3 INTEGRATION WITH IMAGING SYSTEM:

The X-Y positioning system coordinates with the USB camera and NeoPixel lighting array to create a fully automated scanning workflow:

SCANNING SEQUENCE:

1. System performs home sequence (X and Y axes to zero position)
2. NeoPixel LEDs illuminate to configured RGB values
3. X-Y system moves camera to first card position
4. Camera captures image (480p or 1080p resolution)
5. Arduino sends "next position" command
6. System moves to next card in grid pattern
7. Process repeats for all cards in layout
8. System returns to home position upon completion

3.4 ARDUINO FIRMWARE INTEGRATION:

The X-Y positioning system is controlled via the Arduino Mega 2560 running custom firmware (v4.0+) with the following capabilities:

- G-code command interpretation for motion control
- Real-time position tracking and reporting
- Synchronized LED control during positioning
- Serial communication with Python host (9600 baud)
- Emergency stop functionality
- Position caching for repeated scans

COMMAND PROTOCOL EXAMPLE:

- "MOVE X100 Y150" → Move to position (100mm, 150mm)
- "HOME" → Return all axes to zero position
- "SCAN_GRID 9 5" → Scan 9x5 card grid
- "GET_POS" → Report current X-Y coordinates

3.5 STRUCTURAL FRAME:

- FRAME CONSTRUCTION:
  - Aluminum extrusion frame (2020 or 2040 profile)
  - Dimensions: 500mm (X) x 700mm (Y) x 400mm (Z)
  - Rigid mounting for stepper motors and lead screws
  - Adjustable leveling feet for surface compensation
  - Vibration dampening mounts to reduce motion blur

- CARD SCANNING BED:
  - Flat scanning surface (tempered glass or acrylic)
  - Non-reflective coating to minimize glare
  - Integrated lighting from below (optional backlighting)

- Grid marking overlay for manual card placement
- Removable design for cleaning and maintenance

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PATENT CLAIMS - MECHANICAL SYSTEM ADDITIONS

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CLAIM 11: A precision X-Y axis positioning system comprising:

(a) Dual lead screw assemblies for X and Y axis movement

(b) NEMA 17 stepper motors with microstepping drivers

(c) Linear guide rails with ball bearing carriages

(d) Camera carriage mount with integrated LED lighting

(e) Arduino-based motion control firmware

(f) End-stop limit switches for positional homing

(g) G-code command protocol for position control

CLAIM 12: The positioning system of Claim 11, wherein the system achieves positional accuracy of ±0.1mm through microstepping and software backlash compensation.

CLAIM 13: An automated card scanning sequence utilizing the X-Y positioning system of Claim 11, comprising:

(a) Initialization via home sequence to zero position

(b) Grid-based movement pattern for sequential card capture

(c) Synchronized LED illumination during image acquisition

(d) Position reporting to host computer via serial interface

CLAIM 14: The mechanical frame structure supporting the X-Y axis system, comprising aluminum extrusion frame with vibration dampening and adjustable leveling for optimal scanning performance.

a) Arduino Microcontroller Unit:

- Firmware version 4.0+ with enhanced command protocol
- Commands: 'R' (Red LED), 'G' (Green LED), 'B' (Blue LED), 'W' (White LED), 'O' (LEDs Off), 'S' (Servo position), 'C' (Capture trigger)
- RGB values transmitted as 3-digit integers (000-255)
- Servo positions: 0° (home), 45° (scan position), 90° (hold position)
- Auto-detection across COM1-COM15 ports
- Heartbeat protocol for connection health monitoring

b) Imaging System:

- Primary: DSLR camera (Canon/Nikon compatible)
  * Resolution: 12+ megapixels
  * Focus: Manual with macro lens capability
  * Connection: USB tethering via DirectShow
- Secondary: HD webcam (1080p minimum)
  * Purpose: Real-time preview and backup recognition
  * Frame rate: 30fps minimum
- Calibration system for lens distortion correction

c) Illumination System:

- NeoPixel RGB LED strip (60 LEDs/meter)
- Individual LED addressability
- Color temperature control (2700K-6500K)
- Brightness control (0-100% in 1% increments)
- Preset modes: daylight, studio, UV detection
- Diffusion panel for even illumination

d) Mechanical Positioning:

- Servo-controlled card holder
- Precision: ±0.5° angular accuracy
- Speed: Adjustable 10-500ms per movement
- Safety features: Torque limiting, position feedback

2. SOFTWARE ARCHITECTURE:

AUTOMATED TRADING CARD SCANNING SYSTEM - IMAGING & DATA WORKFLOW

Applicable to: Magic: The Gathering (MTG) & Pokémon Trading Card Game (TCG)

Step 1: Card Detection

IR sensor (Pin 7) detects card presence

Arduino triggers NeoPixel LEDs (Pin 13) for optimal lighting

System readies camera for capture

Step 2: Image Acquisition

DSLR camera (1024x768) or Webcam (1280x720) captures high-res image

DirectShow backend ensures reliable capture

Image stored temporarily in memory buffer

Step 3: Advanced OCR Processing (4-Corner Tesseract)

OpenCV preprocessing: Crop, enhance contrast, normalize brightness

4-corner text extraction: Tesseract OCR targets card-specific regions:

Magic The Gathering:

Top-left: Card name

Top-right: Mana cost

Bottom-left: Power/Toughness

Bottom-right: Set symbol + collector number

Pokémon Cards:

Top-left: Card name

Top-right: HP value

Bottom-left: Set symbol + card number

Bottom-right: Rarity symbol

Text cleaning: Remove artifacts, normalize spacing

Confidence scoring: Each OCR result rated 0-100% accuracy

Step 4: Database Matching

MTG: Queries Scryfall database (100,000+ cards)

Pokémon: Queries Pokémon TCG API database (15,000+ cards)

Fuzzy matching algorithm finds best match (Levenshtein distance ≤3 edits)

Cross-references: name + set + collector number for exact match

Returns: card details, pricing, legality/format, image URI

Step 5: Data Storage

Card added to SQLite inventory database

Records: name, set, quantity, condition, game type (MTG/Pokémon), acquisition date, price

Updates collection statistics in real-time

Step 6: Card Ejection

Motor 1 (Pins 2, 5) activates to eject scanned card

LED pattern changes to indicate completion (green = success, red = error)

System ready for next card (cycle repeats)

Key Innovation: Universal 4-corner Tesseract OCR adapts to MTG and Pokémon card layouts for 98%+ accuracy. Supports fuzzy matching for damaged/foreign cards. Fully automated pipeline from physical card → digital inventory in <3 seconds per card, regardless of game type.

Patent Coverage: Trading card scanning system applicable to any collectible card game with corner-based text layouts (MTG, Pokémon, Yu-Gi-Oh!, etc.).

B2B (BUSINESS-TO-BUSINESS) CAPABILITIES

1. Local Game Store (LGS) Inventory Management

Automated Bulk Scanning: Hardware scanner processes entire collections at 98%+ accuracy, <3 seconds per card

Real-Time Pricing: Scryfall API integration for market-accurate valuations

Inventory Tracking: SQLite database tracks 10+ tables (cards, inventory, sales, marketplace, customers, QuickBooks integration)

Wholesale Authentication: 4-corner Tesseract OCR verifies card legitimacy (name, set, mana cost, P/T for MTG; name, HP, rarity for Pokémon)

Bulk Upload/Export: CSV import/export for store systems, QuickBooks sync for accounting

2. Authentication & Grading Services

Professional Card Verification: Image capture system (DSLR 1024x768 + Webcam 1280x720) with DirectShow backend

Condition Assessment: Database condition tracking (NM, LP, MP, HP, DMG)

Set Completion Analysis: Automated set tracking for collectors and stores

Foil Availability Checker: Cross-references collection against Scryfall foil versions

3. Wholesale Distribution

Peer-to-Peer Marketplace: NEXUS Marketplace connects stores/distributors for bulk trades

Price Intelligence: Market trend analysis, popular card tracking, average pricing data

Transaction History: Complete audit trail for wholesale purchases/sales

Rating System: Trusted seller verification, transaction ratings (1-5 ⭐)

B2C (BUSINESS-TO-CONSUMER) CAPABILITIES

1. Online Marketplace

Card Listings: Browse 100,000+ MTG cards (Scryfall) + 15,000+ Pokémon cards (TCG API)

Search Filters: Name, set, price, condition, seller location

Watchlist Feature: Track favorite cards, price alerts

Buy Now / Make Offer: Direct purchase or negotiation options

Seller Profiles: Ratings, completed sales, revenue tracking

2. Collection Management

Automated Scanning: Home users scan collections with hardware setup

Portfolio Tracking: Real-time collection value, card counts, average values

Import/Export: CSV, Gestic, Untapped.gg deck imports

Set Completion: Track progress toward complete sets

Price Updates: One-click Scryfall price refresh across entire collection

3. Deck Building & Testing

AI-Powered Deck Builder: Multi-format support (Commander, Standard, Modern, Pioneer, Legacy, Vintage, Pauper, Brawl)

Strategy Optimization: 6 archetypes (balanced, aggro, control, combo, midrange, tempo)

Deck Testing Suite: Goldfish testing, combat simulation, mana analysis, meta comparison

Deck Copies Calculator: "How many decks can I build?" from your inventory

Smart Substitutions: AI suggests cards from YOUR collection only

4. Sales & Transaction Tracking

For Sale Listings: Mark cards for sale with custom pricing

Sold Card History: Complete transaction records with buyer info

Sales Reports: Revenue analytics, profit tracking

QuickBooks Integration: Automated accounting sync for hobbyists/small businesses

PATENT-RELEVANT CLAIMS

Universal TCG Platform

MTG + Pokémon TCG + extensible to Yu-Gi-Oh!, other collectible card games

4-corner OCR methodology applicable to any card layout

Database-agnostic matching (Scryfall for MTG, TCG API for Pokémon, custom APIs for future games)

Hardware-Software Integration

X-Y axis mechanical positioning (NEMA 17 steppers, 400mm×600mm travel, ±0.1mm accuracy)

Arduino Mega 2560 automation (IR sensors, NeoPixel LEDs, motor control)

Dual-camera system (DSLR + webcam) with DirectShow backend

6-step automated workflow: Detection → Acquisition → OCR → Matching → Storage → Ejection

Marketplace Innovation

Peer-to-peer trading within single platform (no external marketplaces)

Real-time pricing from multiple APIs

Trusted seller verification system

Transaction escrow/rating hybrid for user trust